PRACTICAL 2:

Aim: Automate student marksheet generation, system information display, Fibonacci and prime number generation, and file management operations using shell scripts to enhance computational efficiency and user interaction.

a) Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.

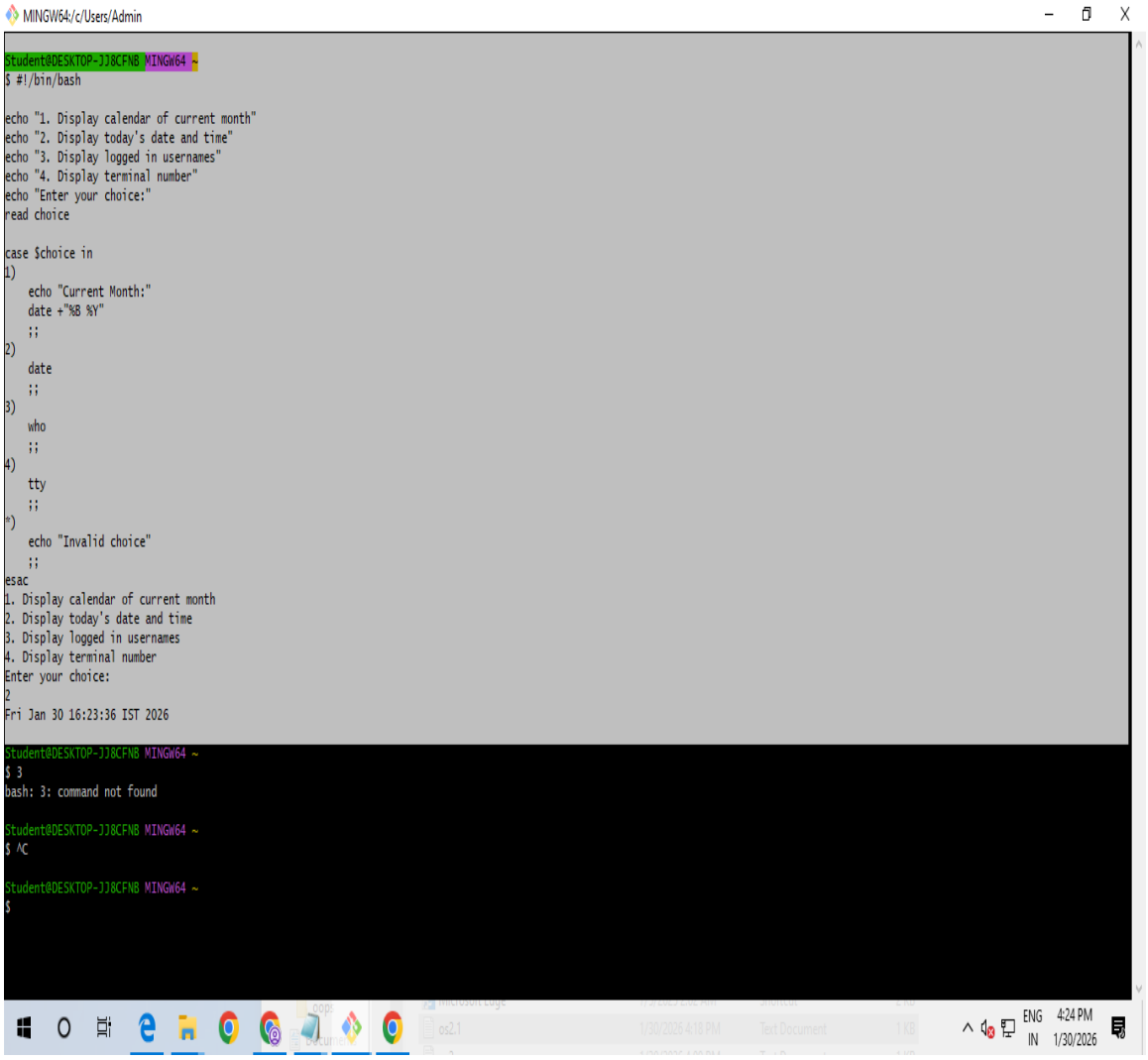
MINGW64:/c/Users/Admin

```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ #!/bin/bash

echo "Enter marks of subject 1:"
read m1
echo "Enter marks of subject 2:"
read m2
echo "Enter marks of subject 3:"
read m3

total=$((m1 + m2 + m3))
percentage=$((total / 3))

echo "Total Marks = $total"
echo "Percentage = $percentage%"

if [ $percentage -ge 60 ]; then
    echo "Class: First Class"
elif [ $percentage -ge 50 ]; then
    echo "Class: Second Class"
elif [ $percentage -ge 40 ]; then
    echo "Class: Pass"
else
    echo "Class: Fail"
fi
Enter marks of subject 1:
67
Enter marks of subject 2:
87
Enter marks of subject 3:
84
Total Marks = 238
Percentage = 79%
Class: First Class

Student@DESKTOP-JJ8CFNB MINGW64 ~
$
```

ENG IN 4:14 PM 1/30/2026

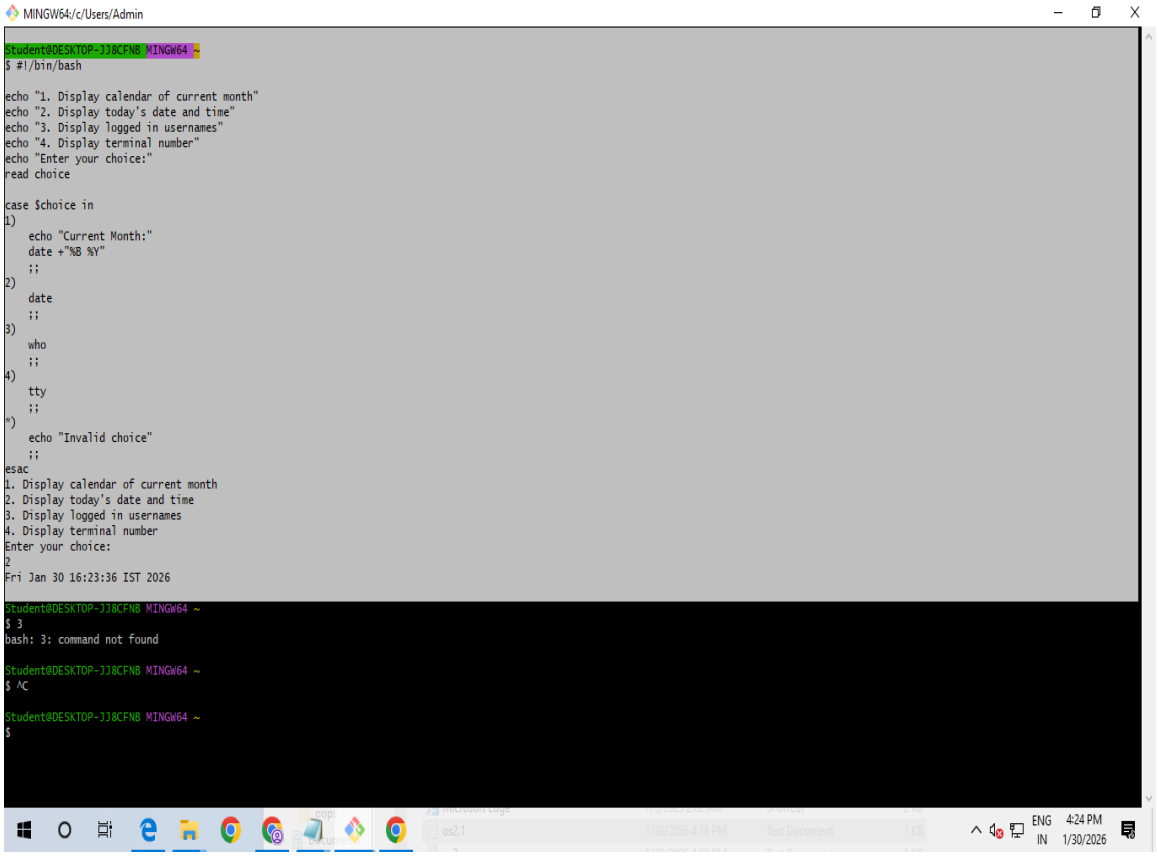
b) Write a menu driven shell script which will print the following menu and execute the given task.

- Display calendar of current month.



```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ #!/bin/bash

echo "1. Display calendar of current month"
echo "2. Display today's date and time"
echo "3. Display logged in usernames"
echo "4. Display terminal number"
echo "Enter your choice:"
read choice

case $choice in
1)
    echo "Current Month:"
    date +"%B %Y"
    ;;
2)
    date
    ;;
3)
    who
    ;;
4)
    tty
    ;;
*)
    echo "Invalid choice"
    ;;
esac
1. Display calendar of current month
2. Display today's date and time
3. Display logged in usernames
4. Display terminal number
Enter your choice:
2
Fri Jan 30 16:23:36 IST 2026

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ 3
bash: 3: command not found

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ ^C

Student@DESKTOP-JJ8CFNB MINGW64 ~
$
```

- Display today’s date and time



c) Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

MINGW64:/c/Users/Admin

```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ #!/bin/bash

echo "Enter number of terms:"
read n

a=0
b=1

echo "Fibonacci Series:"
for ((i=1; i<=n; i++))
do
    echo -n "$a "
    c=$((a + b))
    a=$b
    b=$c
done
echo
Enter number of terms:
7
Fibonacci Series:
0 1 1 2 3 5 8

Student@DESKTOP-JJ8CFNB MINGW64 ~
$
```

d) Write a shell script which will accept a number b and display first n prime numbers as output

MINGW64:/c/Users/Admin

```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ #!/bin/bash

echo "Enter number of prime numbers:"
read n

count=0
num=2

echo "Prime Numbers:"
while [ $count -lt $n ]
do
    flag=1
    for ((i=2; i<=num/2; i++))
    do
        if [ $((num % i)) -eq 0 ]; then
            flag=0
            break
        fi
    done

    if [ $flag -eq 1 ]; then
        echo -n "$num "
        count=$((count + 1))
    fi
    num=$((num + 1))
done
echo
Enter number of prime numbers:
9
Prime Numbers:
2 3 5 7 11 13 17 19 23

Student@DESKTOP-JJ8CFNB MINGW64 ~
$
```

e) Write menu driven program for file handling activity • Creation of file.

MINGW64:/c/Users/Admin

```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ #!/bin/bash

while true
do
    echo "----------------------------------"
    echo "        FILE HANDLING MENU        "
    echo "----------------------------------"
    echo "1. Create a file"
    echo "2. Write content to file"
    echo "3. Append content to file"
    echo "4. Delete file content"
    echo "5. Exit"
    echo "----------------------------------"
    echo "Enter your choice:"
    read choice

    case $choice in
        1)
            echo "Enter file name:"
            read fname
            touch "$fname"
            echo "File created successfully"
            ;;

        2)
            echo "Enter file name:"
            read fname
            echo "Enter content (Press CTRL+D to save):"
            cat > "$fname"
            echo "Content written to file"
            ;;

        3)
            echo "Enter file name:"
            read fname
            echo "Enter content to append (Press CTRL+D to save):"
            cat >> "$fname"
            echo "Content appended to file"
            ;;

        4)
            echo "Enter file name:"
            read fname
            > "$fname"
            echo "File content deleted"
            ;;

        5)
```

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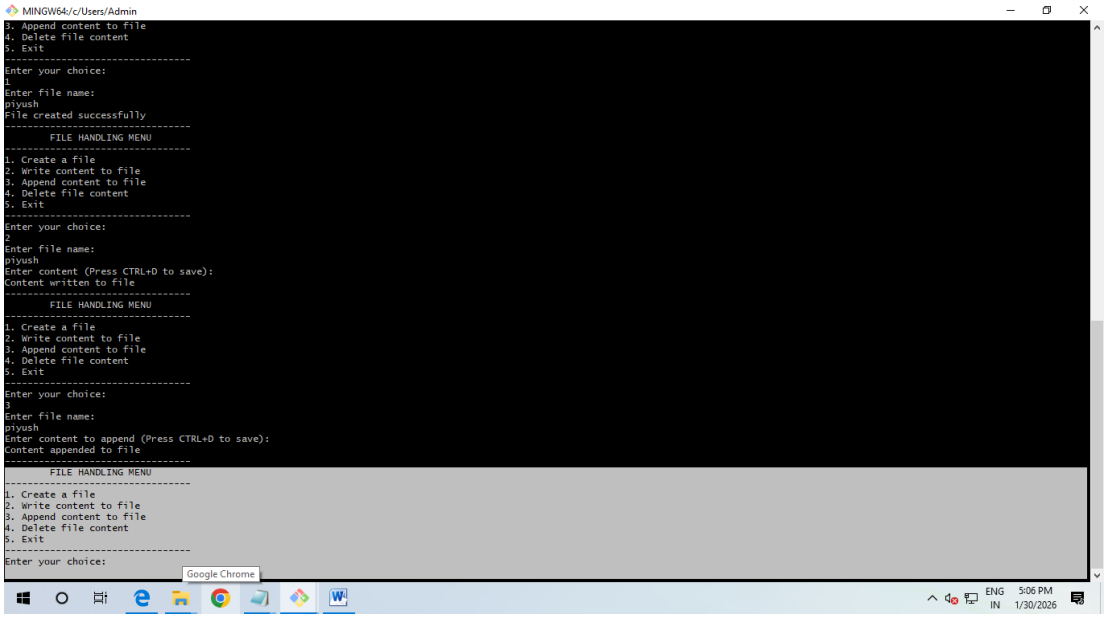
- Write content in the file.